

ElderMeds: A Multimodal Voice Vision Based Intelligent Assistant for Medication Adherence

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Abstract— Medication non-adherence in older adults is influenced by polypharmacy, memory limitations, medication-safety concerns, and changing daily routines. This paper presents ElderMeds, a unified multimodal AI system integrating routine-aware reminders, verified medication intake, patient-specific medication-safety assessment, emotional and cognitive support, risk-awareness, caregiver monitoring, and conversational access to longitudinal records. The system combines mobile and web technologies, computer vision, machine learning, deterministic clinical rules, and a constrained local-LLM dashboard. Evaluation is reported by evidence type rather than as a single system accuracy. On an 860-image held-out test set, YOLO11n pill counting achieved F1=0.9803 and exact-count accuracy=0.9093. An independent frozen 140-response ElderMeds-domain test of the emotion classifier achieved accuracy=0.8143 and macro F1=0.8166, with a 95% bootstrap confidence interval of 0.7461–0.8744. The medication-safety experiment used 528,000 processed FAERS records; Tuned Logistic Regression achieved 86.88% hold-out accuracy, recall=0.9817, F1=0.9180, and ROC-AUC=0.9544. The hybrid medication-safety controller, using 0.6/0.4 rule-ML weighting and 34/36 operating thresholds, achieved 81.55% proxy-calibration accuracy, weighted F1=0.8118, Dangerous-class precision=0.8220, and recall=0.9750, corresponding to a 0.0250 false-negative rate. Independent medical-expert review supported the clinical plausibility of representative medication-safety reasoning and outputs within the intended research scope. Overall, ElderMeds demonstrates the feasibility of coordinated and auditable multimodal integration for adherence verification, patient-specific safety reasoning, supportive engagement, risk-awareness, caregiver monitoring, and retrieval-grounded access through shared longitudinal patient context.

Keywords—elderly care, medication adherence, medication safety, multimodal AI, explainable AI.

I. INTRODUCTION

Medication non-adherence remains a persistent challenge in chronic disease management and is particularly difficult in older adults, where polypharmacy [14], memory impairment, visual or motor limitations, changing routines, adverse-drug concerns, and social or emotional factors may occur simultaneously. Conventional reminder applications mainly address when a medicine should be taken, but they do not necessarily verify what was taken, connect adherence evidence with personalized medication-safety reasoning, or

provide a unified mechanism for caregivers to interrogate cross-module health information.

ElderMeds was designed as one integrated elder-care research system rather than a collection of independent component studies. It combines four interdependent functional streams: (1) intelligent routine-aware medication reminders with the Medication Intake Monitoring and Verification (MIMV) subsystem; (2) real-time medication-safety assessment combining patient-specific clinical rules with population-level pharmacovigilance evidence; (3) a voice-first, context-aware emotional and cognitive engagement component organized around TALK, PLAY, REMEMBER, and REVIEW; and (4) a Unified Conversational Dashboard that connects medication events, safety outputs, emotional-session information, caregiver alerts, disease-risk results, routines, and conversation history through a common data layer.

The research gap addressed by ElderMeds is the fragmentation of elder-care technologies that separately support medication reminders, intake verification, medication safety, emotional and cognitive engagement, risk-awareness, caregiver monitoring, and conversational health access [1]–[8]. ElderMeds addresses this gap through a unified multimodal architecture in which these capabilities operate over shared longitudinal patient and medication context. Its system-level novelty therefore lies in coordinated and auditable integration, cross-module persistence and retrieval, and explicit safety boundaries between probabilistic AI outputs and patient-specific clinical reasoning.

The principal contributions are fourfold. First, ElderMeds provides a unified multimodal architecture connecting adherence verification, medication safety, supportive engagement, risk-awareness, and caregiver interaction through longitudinal context. Second, it implements a verified-intake workflow that combines pill counting, medicine-identity evidence, hand-to-mouth motion, and explicit user confirmation. Third, it introduces an auditable hybrid medication-safety architecture that separates population-level adverse-event ML evidence from patient-specific clinical rules and guardrails. Fourth, it provides a constrained conversational integration layer for user-scoped access to cross-module records. The evaluation therefore

reports subsystem evidence separately and avoids transferring performance claims from one task to another.

II. RELATED WORK AND RESEARCH GAP

A. Medication Adherence and Intake Verification

Mobile medication applications and reminder interventions can improve adherence [1]–[4], but reminder delivery alone does not establish that the correct medicine and quantity were consumed. Computer vision has separately been investigated for pharmaceutical image recognition [5], [6]. ElderMeds adopts these ideas in a verified-intake workflow that combines pill counting, medicine-identity evidence, hand-to-mouth motion, and explicit confirmation rather than treating a single visual signal as proof of ingestion.

B. Medication Safety Decision Support

Clinical decision-support systems commonly use drug knowledge bases, interaction rules, and machine-learning models. ElderMeds emphasizes separation of evidence roles: public adverse-event reports provide a population-level pharmacovigilance signal, whereas allergies, current medicines, interactions, chronic conditions, age, prior reactions, and questionnaire evidence drive patient-specific escalation. This distinction prevents a probability learned from spontaneous public reports from being presented as an individual probability of harm. RxNorm, SIDER, DDInter, WHO ATC, and FAERS are used as medication-knowledge and pharmacovigilance resources [9]–[13].

C. Conversational and Emotional Support

Voice interfaces, conversational health systems, reminiscence applications, and emotion classifiers have substantial precedent [8], [19]. ElderMeds connects these capabilities to shared medication-care context. The emotional and cognitive engagement component is explicitly non-clinical: voice, typed answers, and Quick Replies converge into the same contextual interpretation path; a five-turn adaptive check-in supports activity recommendation; cognitive gameplay adapts difficulty and variety without being treated as diagnosis; and reminiscence topics are retained only with user consent. The dashboard uses local LLM processing for schema-aware querying and grounded response generation, while deterministic validation, user scoping, confirmation gates, and audit logs constrain access to healthcare data.

D. System-Level Gap

The identified gap is not the absence of individual AI techniques but the fragmentation of reminder, medication-safety, emotional-support, risk-awareness, and caregiver-information tools. ElderMeds addresses this gap through shared patient and medication context. Outputs produced by one module are persisted as contextual evidence that can be retrieved by another system interaction without presenting the modules as independent research systems.

III. INTEGRATED SYSTEM ARCHITECTURE

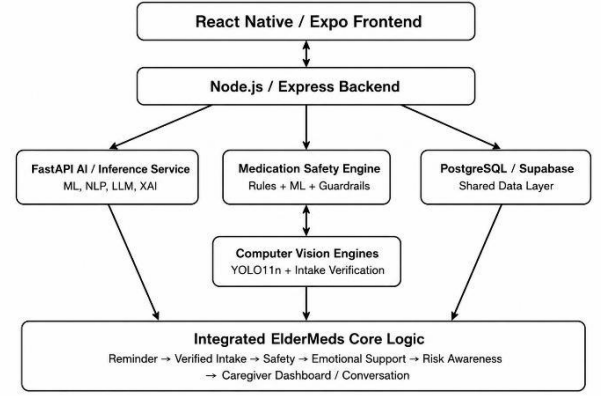


Fig. 1. High-level ElderMeds architecture and cross-module data flow.

The mobile layer is implemented using React Native/Expo and provides medication scheduling, voice interaction, camera capture, safety-result views, emotional-support activities, disease-risk prediction screens, and caregiver/dashboard interfaces. A Node.js/Express backend authenticates requests, orchestrates workflows, validates access, applies deterministic safety logic, and persists system events. FastAPI services host ML, NLP, local-LLM, and XAI inference. PostgreSQL hosted on Supabase stores profiles, routines, medications, adherence events, allergy/reaction histories, emotional sessions, alerts, predictions, conversation history, and action-audit records.

The architecture deliberately separates deterministic and probabilistic responsibilities. Safety-critical writes are confirmation-gated, clinical rules remain independently auditable from the ML branch, and computer-vision outputs are stored as verification evidence. The Unified Conversational Dashboard retrieves shared records through validated backend paths; natural-language requests can be translated into schema-aware queries using a retrieval-augmented generation approach [18], while summary cards use deterministic SQL. This allows reminder status, verified intake, safety outcomes, emotional information, caregiver alerts, and risk-awareness results to be presented through one consistent system context.

Each module persists its outputs as structured records in dedicated PostgreSQL tables rather than exchanging results directly between modules in memory. Verified-intake outcomes, detected-medicine identity/confidence, status events, and stock changes are persisted with the medication record context; hybrid safety scores, fired-rule identifiers, and ML risk factors are written to medicine-check-history and safety-card records; the emotional/cognitive component stores approved longitudinal check-in/session information and consented remembered topics, while raw user-selected reminiscence photos remain local in the current implementation; disease-risk predictions, their input snapshots, and XAI explanations are written to prediction history; and caregiver-facing events are persisted in the shared caregiver-alert layer. The Unified Conversational Dashboard subsequently retrieves cross-module evidence through two validated backend paths rather than reading raw tables directly: deterministic, pre-written SQL for summary cards and a schema-restricted natural-language-to-SQL (NL2SQL) pipeline—constrained by an allowlisted schema digest,

SELECT-only validation, mandatory user-ID parameterization, and row/timeout limits—for open-ended conversational queries. This persist-then-retrieve pattern allows one module's output to become contextual evidence for another interaction without direct in-memory coupling between modules.

IV. METHODOLOGY

A. Intelligent Medication Reminder and Verified Intake

Medication schedules are derived from user routine slots such as breakfast, lunch, dinner, and sleep, with configurable before/after offsets. At intake time, the MIMV workflow captures a palm/tablet image, runs YOLO11n pill detection and counting, compares the detected count with the expected dose, and evaluates medicine identity using colour/shape cues plus metric-learning visual retrieval against ePillID- and Pillbox-derived reference indices. The backend then compares detected identities and counts with the active schedule and can report correct, underdose, overdose, missing, or unexpected medicine outcomes. Motion verification uses MediaPipe hand and face landmarks: wrist elevation, hand-to-mouth proximity, and minimum mouth-area dwell must jointly hold. If MediaPipe landmark extraction fails, for example because of occlusion, poor lighting, or an obscuring camera angle, OpenCV optical flow estimates aggregate upper-frame motion as a coarse fallback proxy; it is therefore a graceful-degradation fallback, not a second independent confirmation signal. A final manual swallowing confirmation is retained because camera motion alone cannot establish physiological swallowing. Only after the sequential capture, count, identity, motion, and manual-confirmation gates pass is the dose recorded as Taken and stock updated.

YOLO11n was trained on 5,894 images and evaluated on a held-out test set of 860 images from a 6,754-image public pill-detection manifest. The 860 test images were excluded from model training and were used only for final performance evaluation. The full manifest contained pill counts 0–6 (456, 687, 1,136, 1,244, 1,192, 946, and 1,093 images, respectively). For the reported intake protocol, detected count is compared with an expected count of two; binary outcomes on the held-out test set were TP=697, TN=135, FP=8, and FN=20, with exact-count accuracy evaluated separately. These measurements therefore represent held-out test performance rather than training performance. Because the training and test images originate from the same public manifest, the 860-image set is described as a held-out test set rather than an independently collected external cohort. Broader external validation would require a separately sourced cohort excluded from all model development, with variation in pill count, morphology, illumination, capture device, background/palm appearance, orientation, overlap, and occlusion.

B. Personalized Medication-Safety Assessment

The medication-safety pipeline consists of conservative medicine normalization, medication-knowledge retrieval, patient-specific clinical-rule evaluation, ML adverse-event prediction, and hybrid decision/explanation generation. Normalization prioritizes trusted aliases and strong local exact matches before RxNorm fallback; unresolved inputs are preserved rather than forced to an unrelated medicine. This

behavior was strengthened after controlled validation exposed incorrect brand-name mappings.

The clinical layer uses a sixteen-rule catalogue (P1–P16), defined in Table I. Each rule specifies its patient-safety logic and base score contribution. The catalogue spans direct and cross-reactive allergy matches, drug–drug interactions, pregnancy-related configured risk, dangerous combination patterns, side-effect overlap, chronic-condition cautions, previous reactions, polypharmacy, age, narrow-therapeutic-index medicines, renal/hepatic impairment interactions, prior-check history, and weak medication-knowledge matches. The updated ML training corpus contains 528,000 processed FAERS records with a binary serious/non-serious adverse-event target. Thirteen report-level features were used, including temporal, reaction, suspect-drug, route, indication, pharmaceutical-class, medication-count, demographic, country, and report-age information. Logistic Regression, Random Forest, and XGBoost [15] were compared using a stratified 80/20 split (422,400 training records and 105,600 held-out records). Logistic Regression provided the strongest overall baseline balance, with accuracy=0.8990, F1=0.9293, and ROC-AUC=0.9648, and was therefore selected for tuning. The final Tuned Logistic Regression used $C=0.01494$, class weights {0:1, 1:3}, and a decision threshold of 0.55. On the held-out set it achieved accuracy=0.8688, precision=0.8621, recall=0.9817, F1=0.9180, and ROC-AUC=0.9544. Its output is treated as a population-level FAERS serious-event signal rather than an individualized probability of harm. The deployed hybrid score is $H=\text{round}(0.6R+0.4M)$, where R is the patient-specific rule score and M is the calibrated ML danger score. The configured user-facing bands are Safe (<34), Warning (34 to <36), and Dangerous (≥ 36). After fixing the selected 0.6/0.4 rule/ML weighting, the 34/36 operating thresholds were chosen through threshold-sensitivity analysis because they provided the preferred safety-oriented performance balance among the evaluated candidate operating points: the configuration retained Dangerous recall of 0.9750 and a Dangerous false-negative rate of 0.0250 while achieving Dangerous precision of 0.8220, with stronger overall class balance than nearby alternatives. The thresholds were therefore selected from empirical proxy-calibration performance rather than assigned arbitrarily. These bands are screening outputs rather than clinical diagnoses.

TABLE I. PATIENT-SPECIFIC CLINICAL RULE CATALOGUE (P1–P16)

Rule	Patient-Safety Logic	Base Contribution
P1	Direct medicine–allergy match	+80 (short-circuit)
P2	Drug-class allergy / cross-reactivity match	+65, severity-adj.
P3	Aspirin/salicylate allergy with another NSAID	+60, severity-adj.
P4	DDInter drug–drug interaction	+10 / +30 / +60
P5	Recorded pregnancy with higher-risk medicine	+70
P6	Dangerous combination (e.g., anticoagulant + NSAID)	+32
P7	SIDER symptom/side-effect overlap	+12
P8	Chronic-condition contraindication/caution	$\approx +10$
P9	Previous reaction evidence	+5
P10	Polypharmacy (≥ 5 current medicines)	+7
P11	Age ≥ 65 years	+5

Rule	Patient-Safety Logic	Base Contribution
P12	Narrow therapeutic-index medicine	+15
P13	Renal impairment + renally cleared medicine	+10
P14	Hepatic impairment + hepatically metabolized medicine	+10
P15	Prior system-check history (supporting context only)	dynamic, capped +45
P16	Weak/absent medication-knowledge match	+12

After defining the catalogue in Table I, the rules are applied according to their configured priority/salience, evidence source, severity, and score contribution. Scores are accumulated and capped at 100, while high-priority score floors and guardrails protect strong patient-specific evidence. P1 has the highest salience and short-circuits lower-priority rules when a direct documented allergy is present.

C. Three-Class Calibration and Evidence Boundaries

The selected hybrid medication-safety operating point demonstrated strong sensitivity to Dangerous proxy cases. With the deployed 0.6 rule/0.4 ML weighting and 34/36 thresholds, evaluation on the 2,374-row proxy-calibration set achieved 81.55% accuracy, macro F1=0.6157, weighted F1=0.8118, Dangerous precision=0.8220, Dangerous recall=0.9750, and Dangerous FNR=0.0250 (Table II). The selected operating point therefore preserved 97.50% Dangerous-class sensitivity while limiting the Dangerous false-negative rate to 2.50%. The three-class hybrid calibration does not use clinician-adjudicated Safe/Warning/Dangerous ground truth. The binary FAERS serious/non-serious target is used only for the pharmacovigilance ML experiment; the three-class calibration labels are constructed proxies according to the predefined calibration procedure. The 34/36 thresholds represent the selected safety-oriented operating point from the threshold-sensitivity comparison and were retained because they preserved very high Dangerous-class sensitivity while improving Dangerous precision and maintaining a stronger balance across the reported aggregate metrics than the nearby candidate thresholds considered. This selection concerns proxy-calibration performance and does not constitute optimization against clinician-adjudicated clinical outcomes. Only 20 of 187 proxy-Warning records were correctly assigned to Warning, showing materially weaker middle-class discrimination. Moreover, the same 2,374-row partition was searched across candidate weight/threshold combinations; therefore, these results represent operating-point calibration rather than performance on an independently locked final test set. Accordingly, the reported accuracy and F1 scores characterize internal proxy-calibration behavior and do not establish clinical safety; independently held-out clinician- or pharmacist-adjudicated validation remains necessary.

TABLE II. HYBRID MEDICATION-SAFETY OPERATING-POINT COMPARISON

Configuration	N	Accuracy	Macro F1	Weighted F1	Dangerous Prec.	Dangerous Recall	FNR
0.6/0.4, 34/36	2,374	0.8155	0.6157	0.8118	0.8220	0.9750	0.0250

The final deployed/proxy-calibrated medication-safety controller uses 0.6/0.4 rule/ML weighting with 34/36 Warning/Dangerous thresholds. In addition to the 2,374-row proxy calibration/evaluation, the medication-safety framework and representative outputs were independently reviewed by medical practitioners. The review covered representative patient-medication scenarios, identified clinical risk factors, the P1–P16 rule framework and risk-scoring rationale, the hybrid rule–ML decision process, and resulting Safe, Warning, and Dangerous classifications with their explanations. The reviewed classifications and explanations were considered clinically reasonable/appropriate within the intended medication-safety screening and decision-support scope. This evidence is reported as independent clinical expert review and clinician-informed feasibility/plausibility evidence, not as a statistically powered clinical validation study or proof of population-level generalizability. Accordingly, a larger independently locked, blinded, clinician/pharmacist-adjudicated evaluation remains necessary. The medication-safety evidence therefore separates the 528,000-record FAERS serious-event ML experiment, the 2,374-row proxy calibration/evaluation, and the independent medical-expert review, without presenting the proxy evaluation or expert review as definitive clinical validation of the complete controller.

D. Emotional and Cognitive Engagement

The emotional and cognitive engagement component is a voice-first, context-aware, non-clinical wellness-support workflow organized around TALK, PLAY, REMEMBER, and REVIEW. TALK uses an exactly five-turn adaptive micro-conversation; voice answers are transcribed and reviewed before submission, while Quick Replies and typed answers enter the same interpretation pipeline. The calibrated emotion model is used with contextual evidence to provide brief supportive acknowledgements and activity selection rather than diagnosis. PLAY provides dynamic cognitive activities with Gentle and A Little More Challenge difficulty, preference signals, recency penalties, and variety controls; gameplay performance is not interpreted as cognitive impairment and does not trigger caregiver alerts. REMEMBER supports a Life Book, Good Deeds/Pin Potha, and consent-based Remembered Topics; selected photos are handled locally and are not used for facial recognition or identity inference. REVIEW presents non-clinical wellness/engagement history. For quantitative validation, the selected MiniLM-based emotion classifier was evaluated on an independent frozen ElderMeds-domain test set containing 140 responses balanced across seven classes—Happiness, Sadness, Loneliness, Anxiety, Anger, Cognitive Fog, and Neutral—with 20 responses per class. The domain test set was excluded from model training, model selection, calibration, and threshold tuning. The classifier achieved accuracy=0.8143 and macro F1=0.8166. Uncertainty was estimated using 10,000 non-parametric bootstrap resamples, yielding a 95% confidence interval of 0.7461-0.8744 for

macro F1. These results provide independent domain-level validation within the ElderMeds evaluation protocol, while the limited sample size motivates further validation using larger externally collected older-adult response datasets.

E. Unified Conversational Dashboard and Risk Awareness

The Unified Conversational Dashboard is the system-level integration mechanism. It aggregates persisted records from medication, adherence, safety, emotional, routine, alert, conversation, and prediction tables rather than operating as an isolated chatbot. Summary cards use deterministic SQL. For open-ended questions, the backend routes the user request through a schema-restricted NL2SQL path: generated SQL is checked against an allowlisted schema, restricted to SELECT, parameterized with the authenticated user ID, and constrained by row and timeout limits. After safe execution, sensitive identifying fields are redacted. The original question, safely executed SQL, declared intent, user role, and retrieved rows are then formatted as grounding context for the local Llama 3 response generator [18]. The model is constrained to explain only those retrieved records and returns an explicit no-matching-records response when retrieval is empty. Retrieval metadata, including the consulted tables/columns and returned-row count, is retained for the optional grounding explanation. Data-modifying conversational actions use a separate confirmation-gated, allowlisted backend path and are recorded in the action audit log.

Beyond read-only querying, an agentic action layer extends the same conversational surface to confirmation-gated create, update, and delete operations across medication, routine, caregiver-alert, and stock-management records. A deterministic parser converts a request such as adjusting stock or marking a dose taken into a structured action object identifying the target module, operation, and payload; the proposed change is summarized in plain language and executed only after explicit user confirmation, with cancellation discarding the pending action before any write occurs. Every executed action is written to a dedicated audit log recording the before/after values and the confirming user, and the returned response includes a navigation directive that brings the user to the affected screen with the changed record highlighted.

A separate Trusted Health-Advice retrieval-augmented generation layer answers guideline-style “what should I do” questions distinctly from the record-grounded query path above. It retrieves curated prevention and management guidance drawn from WHO, CDC, American Heart Association, and Sri Lanka NCD-unit sources, ranks candidate passages against the user’s own latest disease-risk result using a weighted similarity-and-term-overlap score, and constrains the local LLM to generate a short answer from only the retrieved guidance and the user’s prediction context, under a prompt that forbids diagnosis or medication instructions. A deterministic fallback answer, built directly from the retrieved guidance topics and the recorded risk level, is returned if the LLM is unavailable, and the assistant declines outright when no reliable guidance exists for a given question.

For terminological clarity, this paper uses disease-risk prediction to denote the trained machine-learning classification output produced for diabetes, stroke, and hypertension, and risk-awareness to denote the intended, non-diagnostic use of that output for user information rather than

clinical decision-making; the two terms are used in this consistent sense throughout the paper, and the unqualified term “risk prediction” is avoided elsewhere to prevent ambiguity. Four individual candidates—Logistic Regression, Decision Tree, Random Forest, and XGBoost [15]—together with a soft-voting ensemble of the first three, were trained and compared for each of the three disease-risk experiments. The ensemble was retained only where it matched or exceeded the strongest individual candidate on every prioritized metric; across all three diseases it did not clear that bar, so the best-performing individual algorithm was deployed for each disease rather than the ensemble. Diabetes and stroke prioritize recall before secondary metrics, while hypertension uses balanced accuracy to reduce majority-class selection bias. These outputs are framed as risk-awareness rather than diagnosis. Because the hypertension experiment showed weak discrimination, it is treated as a limitation rather than evidence of validated predictive performance.

V. EVALUATION DESIGN

Evaluation was organized by evidence type because the unified system contains heterogeneous AI and rule-based functions that cannot be represented by one meaningful accuracy value. Table III summarizes completed quantitative and controlled evaluations. The medication-safety evaluation separates the 528,000-record FAERS serious-event ML experiment from the 2,374-row proxy-labelled hybrid-controller calibration/evaluation. For pill counting, the 5,894-image training subset and the 860-image held-out test subset are explicitly separated; the test images were not used for model training. Because both subsets originate from the same public manifest, this constitutes held-out test evaluation rather than independently sourced external validation. Accordingly, the manuscript does not claim external generalizability from the 860-image result. A larger independently sourced pill-counting test set, collected across different capture devices, illumination conditions, backgrounds, pill morphologies, orientations, overlaps, occlusions, and count ranges, is specified as the required next validation stage; it must remain excluded from training, tuning, and model selection and be evaluated with the frozen YOLO11n model. This separation aligns each metric with the evidence from which it was obtained and prevents training, proxy, or software-regression results from being generalized beyond their supported scope.

TABLE III. EVALUATION OVERVIEW BY SUBSYSTEM

Subsystem	Evaluation Set	Primary Purpose
YOLO11n pill counting	6,754-image manifest; 5,894 train / 860 held-out test	Held-out detection and exact-count performance evaluation
Emotion classifier	140 frozen domain responses; 7 balanced classes (20/class)	Independent domain evaluation using accuracy, macro-F1, and bootstrap confidence intervals
Medication-safety ML	528,000 FAERS records; 422,400 train / 105,600 held-out test	Binary Serious vs. Non-serious adverse-event classification
Hybrid safety controller	2,374-row proxy-labelled calibration/evaluation set	Hybrid rule-ML operating-point evaluation

Subsystem	Evaluation Set	Primary Purpose
Disease-risk models	Task-specific public datasets	Task-specific predictive evaluation with explicitly bounded risk-awareness claims

VI. RESULTS

A. Pill Counting

TABLE IV. PILL-COUNTING RESULTS

Metric	Value
Accuracy	0.9674
Precision	0.9887
Recall / Sensitivity	0.9721
F1-score	0.9803
ROC-AUC	0.9133
Specificity	0.9441
Exact Count Accuracy	0.9093
Confusion matrix	TP=697, TN=135, FP=8, FN=20

On the held-out 860-image test set, the detector achieved high precision and recall (Table IV), while exact-count accuracy was lower than binary detection accuracy. The test images were excluded from model training, so the reported values characterize performance on unseen held-out examples from the public pill-detection manifest. Precision was 0.9887, recall was 0.9721, F1-score was 0.9803, and exact-count accuracy was 0.9093. Because the held-out test set originates from the same source manifest as the training subset, these results should not be interpreted as independently sourced external validation. Broader generalizability should be assessed on a separately collected external cohort spanning additional pill morphologies, lighting conditions, devices, backgrounds/palm appearances, orientations, overlaps, and occlusions.

B. Emotion Classification

TABLE V. EMOTION CLASSIFICATION RESULTS

Metric	Value
Domain test responses	140
Accuracy	0.8143
Macro precision	0.8278
Macro recall	0.8143
Macro F1-score	0.8166
Macro OVR specificity	0.9690
Class distribution	7 classes; 20 responses/class
Macro F1 95% CI	0.7461-0.8744 (10,000 bootstrap resamples)

The selected MiniLM classifier was evaluated on an independent frozen ElderMeds-domain test set of 140 responses, balanced across seven classes (20/class) and excluded from training, model selection, calibration, and threshold tuning. As shown in Table V, accuracy was 0.8143 and macro F1=0.8166. Using 10,000 non-parametric bootstrap resamples, the 95% CI for macro F1 was 0.7461-0.8744. This provides independent domain-level validation, although larger externally collected older-adult datasets remain necessary for broader generalizability.

C. Medication-Safety ML and Hybrid Controller

TABLE VI. MEDICATION-SAFETY ML MODEL COMPARISON

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.8990	0.9751	0.8877	0.9293	0.9648
Random Forest	0.8857	0.9075	0.9434	0.9251	0.9463
XGBoost	0.8893	0.9149	0.9394	0.9270	0.9475

As shown in Table VI, among the baseline models, Logistic Regression achieved the strongest overall balance, with 89.90% hold-out accuracy, F1=0.9293, and ROC-AUC=0.9648; Random Forest achieved the highest baseline serious-event recall (0.9434). Logistic Regression was therefore selected for tuning rather than XGBoost. The final Tuned Logistic Regression (C=0.01494, class weights {0:1, 1:3}, threshold=0.55) achieved 86.88% accuracy, precision=0.8621, recall=0.9817, F1=0.9180, and ROC-AUC=0.9544 on the 105,600-record stratified hold-out set. These values characterize binary serious/non-serious classification within the processed FAERS dataset; they are not Dangerous-class clinical-safety metrics and do not estimate an individual patient’s causal probability of harm.

TABLE VII. HYBRID MEDICATION-SAFETY PROXY CALIBRATION RESULTS

Controller Metric	Value
Accuracy	0.8155
Macro F1	0.6157
Weighted F1	0.8118
Dangerous precision	0.8220
Dangerous recall	0.9750
Dangerous FNR	0.0250
Warning precision	0.2140
Warning recall	0.1070
Safe precision	0.7630
Safe recall	0.7920

Table VII reports the final proxy-calibrated controller performance using the selected 0.6/0.4 rule/ML weighting and the 34/36 Safe-Warning-Dangerous thresholds. Accuracy was 0.8155, macro F1=0.6157, and weighted F1=0.8118. The Dangerous class achieved precision=0.8220 and recall=0.9750 (FNR=0.0250). This result demonstrates that the selected operating point preserved high sensitivity to Dangerous proxy cases, correctly identifying 97.50% of this class while limiting the corresponding false-negative rate to 2.50%, consistent with the safety-oriented objective of minimizing missed high-risk cases. The Warning class achieved precision=0.2140 and recall=0.1070, reflecting the narrow warning band. The Safe class achieved precision=0.7630 and recall=0.7920. These results are proxy calibration/evaluation evidence and do not establish clinical reliability or generalizability; independent clinician-adjudicated validation remains required. In addition to these quantitative proxy results, representative medication-safety scenarios, clinical risk factors, rule reasoning, hybrid decision logic, and Safe/Warning/Dangerous outputs were reviewed by medical practitioners, who considered the reviewed outputs clinically reasonable/appropriate within the intended research scope. This clinician review supports feasibility and clinical plausibility, but does not replace a larger blinded clinician-adjudicated validation cohort.

D. Disease-Risk Experiments

TABLE VIII. DISEASE-RISK PREDICTION RESULTS

Risk Model	Algorithm	Accuracy	Precision	Recall	F1	ROC-AUC
Diabetes	XGBoost	0.7427	0.7195	0.7954	0.7556	0.8204
Stroke	Logistic Regression	0.7466	0.1384	0.8000	0.2360	0.8426
Hypertension	Decision Tree	0.5205	0.7226	0.5405	0.6184	0.5031

Diabetes and stroke achieved ROC-AUC=0.8204 and 0.8426, respectively (Table VIII). Stroke used a stratified split on 5,110 records with only 249 positives (4.87%); recall-prioritized Logistic Regression achieved recall=0.8000 but precision=0.1384 and F1=0.2360, indicating a substantial false-positive burden. Hypertension used 174,982 records with 125,781 positives (71.88%) and a source label with weak agreement to a BP-derived clinical indicator. Its Decision Tree ROC-AUC=0.5031 indicates chance-level discrimination; saved-model metadata similarly reports ROC-AUC=0.4955 and balanced accuracy=0.5000. Stroke and hypertension are therefore not presented as reliable risk-awareness capabilities. All disease-risk results remain preliminary pending representative independent external validation.

VII. DISCUSSION

A. System-Level Contribution

ElderMeds' system-level contribution is coordinated operation across shared context: verified intake combines count, identity, motion, and confirmation; medication safety combines patient-specific rules with population-level ML evidence; emotional/cognitive support provides non-diagnostic engagement; and caregiver alerts and disease-risk outputs are persisted for permitted retrieval by the Unified Conversational Dashboard. The value of this architecture lies not only in the individual predictive components but in their coordinated operation. Medication adherence evidence, patient-specific safety reasoning, supportive engagement, caregiver information, and risk-awareness outputs can be persisted and subsequently retrieved within the same user-scoped context. This enables ElderMeds to connect multimodal observations with longitudinal information while preserving explicit boundaries between deterministic safety logic, probabilistic ML evidence, and conversational generation.

The heterogeneous evaluations are intentionally reported separately because pill counting, emotion classification, pharmacovigilance classification, safety control, and disease-risk prediction have different targets, evidence sources, and failure costs; no single component metric is treated as whole-system validation.

B. Safety, Privacy, and Explainability

ElderMeds is a research prototype and not an autonomous diagnostic or prescribing system. Medication-safety outputs are decision-support signals; disease-risk outputs are awareness aids; emotional classifications are non-diagnostic; and verified intake retains explicit user confirmation. JWT authentication, user-scoped database operations, allowlisted

conversational actions, confirmation gates, and audit logs reduce unintended modification. Local LLM processing limits unnecessary external transmission of health data. Explainability is provided through rule identifiers and evidence sources for medication safety, feature-contribution summaries for risk models [16], [17], and retrieval-grounding metadata for conversational answers. Both the feature-contribution summaries and the retrieval-grounding metadata are rendered as an optional, evidence-constrained plain-language explanation exposed to the user on request, so the natural-language rendering step cannot introduce a claim beyond the underlying evidence; the same evidence-only constraint governs the trusted-advice layer, which is restricted to retrieved guidance and the user's own recorded risk result and explicitly excludes diagnostic or prescriptive language.

C. Validation Scope and Evidence Boundaries

Validation claims are bounded by subsystem evidence. Pill counting was trained on 5,894 images and evaluated on a separate 860-image held-out test set that was not used for model training; because both subsets originate from the same public manifest, the result is reported as held-out test performance rather than independently sourced external validation. To directly address the remaining external-validity requirement, future validation is predefined as evaluation of the frozen YOLO11n model on a larger independently sourced test cohort spanning different devices, lighting, backgrounds, pill appearances, orientations, overlap/occlusion, and count ranges, with precision, recall, F1, and exact-count accuracy reported without retraining on that cohort. Emotion classification uses an independent frozen 140-response domain set balanced across seven classes, with macro F1=0.8166 and 95% bootstrap CI=0.7461–0.8744. Medication-safety evidence separates the 528,000-record FAERS serious-event binary ML experiment, the 2,374-row proxy-label calibration/evaluation, and independent medical-expert review of the P1–P16 framework and representative Safe/Warning/Dangerous outputs. Proxy labels are not clinician-adjudicated ground truth, and the expert review is treated as clinician-informed feasibility/clinical-plausibility evidence rather than statistically powered proof of generalizability. Disease-risk evidence is likewise bounded: stroke precision is 0.1384 despite recall=0.8000, and hypertension discrimination is near random (ROC-AUC=0.5031; saved artifact 0.4955). These boundaries prevent unsupported clinical, external-validation, or generalizability claims.

VIII. CONCLUSION

ElderMeds demonstrates the technical feasibility and research value of a unified multimodal architecture that integrates medication-adherence monitoring, patient-specific medication-safety reasoning, non-diagnostic emotional and cognitive support, risk-awareness, caregiver monitoring, and retrieval-grounded conversational access through shared longitudinal patient context. In particular, the medication-safety subsystem demonstrates how population-level FAERS pharmacovigilance evidence can complement, rather than replace, explicit patient-specific P1-P16 clinical rules and safety guardrails. The deployed hybrid controller uses the empirically selected 0.6/0.4 rule/ML weighting and 34/36 operating thresholds; the latter were selected through proxy

threshold-sensitivity analysis for their preferred safety-oriented performance balance, including Dangerous recall=0.9750, precision=0.8220, and FNR=0.0250, rather than being assigned arbitrarily. These proxy results remain distinct from clinical validation. Future work will prioritize independently sourced pill-counting evaluation, larger externally collected older-adult emotion datasets, and a larger blinded clinician/pharmacist-adjudicated medication-safety study to determine generalizability beyond the present research setting. Disease-risk outputs, particularly stroke and hypertension, will likewise require stronger independent validation before reliability claims are warranted.

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